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Summary

I am a Ph.D student at University of Hawai'i at Mānoa. My research focuses on advancing next-generation wireless networks, with primary focus on O-RAN architecture, integrated sensing and communication (ISAC), and network security. Looking ahead, I aim to advance intelligent, secure, and adaptable 6G/Next-G systems to enable applications in localization and spatial sensing, autonomous networking, IoT connectivity, and resilient cyber-physical infrastructures. I was awarded *1st Place at the ECE Research Scholarship Expo – Spring 2024* for my work on federated RF fingerprinting in O-RAN architectures. I completed my graduate studies in *M.S. Applied Artificial Intelligence* from *Stevens Institute of Technology*, and undergraduate studies in *Computer Engineering* at *Rajiv Gandhi Institute of Technology (Mumbai University)*.

Education

University of Hawaii at Mānoa

Ph.D. in Electrical and Computer Engineering

Jan 2026 –

Honolulu, HI

Stevens Institute of Technology

Master of Science in Applied Artificial Intelligence

Sept 2023 – May 2025

Hoboken, NJ

Mumbai University

Bachelor of Engineering in Computer Engineering

Aug 2019 – May 2023

Mumbai, MH

Research Experience

University of Hawaii at Mānoa

Research Assistant

Jan 2026 - Present

Honolulu, HI

Advisor: Prof. Xiaochan Xue

Project 1: Open Radio Access Network (O-RAN) Deployment

Jan 2026 - Present

- Deployed and configured an O-RAN-based 5G testbed using open-source RAN components, near-RT RIC, xApps, and supporting Kubernetes infrastructure.
- Integrated monitoring and visualization tools such as Grafana, InfluxDB, and Telegraf to collect and analyze RAN performance metrics.
- Worked on E2 interface/xApp experimentation to study programmable RAN control, network observability, and intelligent radio resource management.

Project 2: Fake Base Stations in 5G

Jan 2026 - Present

- Investigated 5G fake base station attack surfaces, focusing on early control-plane signaling and vulnerabilities in UE-gNB-core network interactions.
- Analyzed NGAP/N2 signaling behavior to study how fake or compromised gNBs could trigger UE disconnection, denial-of-service, or context manipulation.
- Built an experimental setup using SDR-based gNB/UE components and packet analysis tools to examine message crafting, replay, and detection opportunities.

Stevens Institute of Technology

Graduate Research Assistant

Sept 2024 – Dec 2025

Hoboken, NJ

Advisor: Prof. Shucheng Yu

Thesis: Contactless Respiratory Sensing using mmWave 5G NR and FMCW Radar

Sept 2024 - Jun 2025

- Developed a dual-modality contactless respiration monitoring system using 28 GHz 5G NR and 2 MHz FMCW radar within an ISAC framework, leveraging USRP-2974 SDRs and phased array antennas.
- Built real-time signal acquisition and preprocessing pipelines; extracted respiration features from CSI and radar returns under indoor LoS conditions.
- Boosted model robustness through synthetic augmentation and achieved 98% accuracy across four breathing patterns using a 1D CNN trained on multi-modal features.

Project 1: Federated Learning for RF Fingerprinting for Device Authentication

Jan 2024 - Jun 2024

- Implemented federated learning for RF device fingerprinting on the Open RAN (O-RAN) architecture; deployed a custom xApp on the Near-RT RIC for privacy-preserving, distributed training across base stations.

- Simulated and validated performance on the POWDER testbed using X310/B210 USRPs, achieving 99.75% classification accuracy in real-time 5G O-RAN conditions.
- Awarded 1st Prize at ECE Research Expo Spring 2024 for innovation in secure and scalable wireless edge intelligence.

Project 2: Deep Learning-Based Network Slicing for V2X Communication

June 2024 - Aug 2024

- Developed a deep learning-based network slicing predictor xApp to classify V2X sessions into Low Latency, High Bandwidth, and General slices for dynamic QoS management.
- Trained on Berlin V2X dataset with engineered thresholds, achieving 92% prediction accuracy and enabling adaptive resource allocation via Near-RT RIC.
- Validated performance on an O-RAN testbed with simulated RSUs and UEs, replicating real-world mobility and traffic scenarios.

Publications

- P.1 X. Xue, S. Yu, **S. Parkar** and J. Li, “Waveform Design for ISAC: Trends and Future Directions”, IEEE ICNC 2026 - IEEE International Conference on Computing, Networking and Communications, Maui, Hawaii.
- P.2 X. Xue, **S. Parkar**, S. Yu and Y. Zheng, “AI-Assisted Composite ISAC for mmWave Respiration Pattern Recognition”, IEEE AIOt 2025 - IEEE Annual Congress on Artificial Intelligence of Things, Osaka, Japan
- P.3 X. Xue, S. Yu, **S. Parkar** and Y. Zheng, “ROISD: RIS and O-RAN Assisted Intelligent Sensing for UAV Detection”, IEEE AIOt 2025 - IEEE Annual Congress on Artificial Intelligence of Things, Osaka, Japan.
- P.4 C.-H. Yu, T. Yang, **S. Parkar**, H. Cai, Y. Pan, S. Wu, X. Xue, and Y. Zheng, “Programmable Transmitarray-Assisted Near-Field ISAC for In-Cabin Occupancy Sensing,” in 2026 IEEE 1st Annual Integrated Sensing and Communication Conference (ISAC 2026), Lisbon, Portugal, Nov. 2026.

Honors/ Awards

- 1st Place – ECE Research Scholarship Spring’24 June 2024
- Provost Masters Fellowship Sept 2023

Presentations/ Workshops

Presentations:

- P.1 S. Parkar, X. Xue, S. Yu, ”Federated Learning for RF Device Fingerprinting Over Open Radio Access Networks”, 1st Symposium on Emerging Topics in Networks, Systems, and Cybersecurity, Stevens Institute of Technology, NJ, USA, August 2024. (*Poster Presentation*)

Workshops:

- W.1 NSF CyberTraining: O-RAN-Based Cyberinfrastructure for Future-Generation Wireless Communication and Sensing, Yeshiva University, NY, USA, June 2026. *Invited for Talk on AI-RAN Application*
- W.2 NSF CyberTraining: O-RAN-Based Cyberinfrastructure for Future-Generation Wireless Communication and Sensing, Stevens Institute of Technology, NJ, USA, June 2025. *Invited for instructions on hands-on SDR training*

Graduate Student Mentor

- x96 Vertically Integrated Project/ AFRL SDR Challenge** **Sept 2026**
- Leading an undergraduate research team developing a real-time wireless digital-twin and mapping platform integrating mobile robotics, environmental sensing, and 5G/O-RAN infrastructure.

JESSE - Summer 2026

June - July 2026

- Mentored a team of high-school students in 5G and O-RAN research, guiding the development of an O-RAN-integrated robotic platform with remote camera and navigation capabilities.

AFRL SDR Challenge

Sept 2025 – May 2026

- Led a cross-institutional research team exploring RIS-assisted O-RAN approaches for UAV detection as part of the AFRL SDR Challenge.

Academic Services

- TPC Reviewer IEEE HealthCom 2026: Papers & Posters
- Reviewer: IEEE WCIHS 2026
- Reviewer: UHM GSO Academic Enrichment Awards – Spring 2026
- Volunteer: 1st iCNS/ECE Symposium on AI Research and Innovations (SIT) – Spring 2025

Professional Experience

Stevens Institute of Technology

Jan 2025 – May 2025

Graduate Student Grader – Course: AAI-551

Hoboken, NJ

- Evaluated weekly programming labs, homework, and exams, applying a detailed rubric covering Python syntax, OOP, and data structure fundamentals.
- Held office hours to debug code, clarify lecture material, and guide best practices, resolving student questions.
- Assisted the instructor to refine grading rubrics, develop sample solutions, and maintain grade records in the Canvas LMS for transparent, consistent assessment.

Line Leverage

May 2024 – Dec 2024

Machine Learning Intern

Staten Island, NY

- Sourced and curated statistical data on NBA teams and players.
- Analyzed statistical factors to identify key features influencing team performance.
- Developed and implemented machine learning models to predict match outcomes based on historical performance, enhancing risk management and optimizing betting strategies.

Technical Skills

Programming Languages: Python, C, C++, MATLAB

Operating Systems: Windows, Linux, Unix

Technologies/Frameworks: TensorFlow, NumPy, pandas, scikit-learn, Matplotlib, Docker, Kubernetes, GNU-Radio, srsRAN.

Algorithms: Machine Learning, Deep Learning, Natural Language Processing (NLP), Data Engineering, Machine Vision, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN).

Relevant Coursework

Data Acquisition/Modeling/Analysis, Applied Machine Learning, Probability & Stochastic Processes, Engineering Programming: Python, Data Acquisition: Deep Learning, Pattern Recognition & Classification, Practicum in Applied Artificial Intelligence, Applied Data Structures & Algorithms, Engineering Programming: C++, Engineering Mathematics I–IV, Engineering Mechanics, Basic Electrical Engineering, Engineering Graphics, C Programming, Discrete Structures and Graph Theory, Data Structure, Digital Logic and Computer Architecture, Computer Graphics, Analysis of Algorithm, Database Management System, Operating System, Microprocessor, Theoretical Computer Science, Software Engineering, Computer Network, Data Warehousing and Mining, System Programming and Compiler Construction, Cryptography and System Security, Mobile Computing, Artificial Intelligence, Internet of Things, Machine Learning, Big Data Analytics, Machine Vision, Blockchain, Distributed Computing, Applied Data Science